Standard Discrete Distributions: The Discrete Uniform

We begin our exploration of standard distributions with the most fundamental concept of fairness: a system where every outcome in a finite set is equally likely.

The Discrete Uniform Distribution

Let a random variable X take on any integer value in a contiguous range from a to b inclusive.

- The total number of possible outcomes is $N = b - a + 1$.
- Because no outcome is favored, the probability is distributed evenly across the entire support.

Probability Mass Function (PMF)

We write $X \sim \text{DiscreteUniform}(a, b)$. The PMF is:

$$P(X = k) = \frac{1}{N} = \frac{1}{b - a + 1}, \quad \text{for } k = a, a + 1, \ldots, b$$

Discrete Uniform: Expectation and Variance

1. Expectation (The Midpoint)

Using the definition of expectation and factoring out the constant probability:

$$E[X] = \sum_{k=a}^{b} k \frac{1}{N} = \frac{1}{N} \sum_{k=a}^{b} k$$

The sum of this arithmetic sequence is the average of the first and last terms multiplied by the total number of terms (N):

$$E[X] = \frac{1}{N} \left(N \frac{a+b}{2} \right) = \frac{a+b}{2}$$

Discrete Uniform Variance

Calculating the sum of squares directly from a to b is messy. Instead, we define a shifted variable $Y = X - a + 1$.
Y now takes values $1, 2, \ldots, N$. Because shifting a distribution does not change its spread, $Var(X) = Var(Y)$.

1. Find the Second Moment $E[Y^2]$

Using the standard formula for the sum of consecutive squares:

$$E[Y^2] = \sum_{k=1}^{N} k^2 \frac{1}{N} = \frac{1}{N}\left[\frac{N(N+1)(2N+1)}{6}\right] = \frac{(N+1)(2N+1)}{6}$$

2. Calculate Final Variance

$$Var(Y) = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2$$

$$\implies Var(Y) = \frac{N+1}{2}\left[\frac{N-1}{6}\right] = \frac{N^2-1}{12} \quad \blacksquare$$

The Continuous Uniform Distribution

We begin with the continuous equivalent of perfect fairness.

The Definition

A random variable $X \sim \text{Uniform}(a, b)$ is equally likely to take any real value in the continuous interval $[a, b]$. Because the density is evenly spread out, the PDF is a perfectly flat, horizontal line.

The PDF Formula

To ensure the total area under the curve equals exactly 1 (a rectangle of width $b - a$), the height must be $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Continuous Uniform: Expected Value

Let $X \sim \text{Uniform}(a, b)$ with PDF $f(x) = \frac{1}{b-a}$. We find the expected value via integration:

$$\begin{aligned} E[X] &= \int_a^b x f(x) dx \\ &= \int_a^b x \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right) \end{aligned}$$

Factoring the difference of squares in the numerator, $b^2 - a^2 = (b-a)(b+a)$:

$$E[X] = \frac{1}{b-a} \frac{(b-a)(b+a)}{2} = \frac{a+b}{2} \quad \blacksquare$$

Continuous Uniform: Variance

First, we calculate the second moment, $E[X^2]$:

$$E[X^2] = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$
$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2+ab+b^2)}{3(b-a)} = \frac{a^2+ab+b^2}{3}$$

Now, apply the variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$Var(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$
$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$

Finding a common denominator of 12:

$$Var(X) = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$
$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12} \quad \blacksquare$$

Generating Random Numbers

Computers are inherently deterministic, meaning they cannot generate truly random numbers. Instead, we rely on algorithms called **Pseudo-Random Number Generators (PRNGs)** to create sequences $U_1, U_2, \ldots$ that behave like standard uniforms, Uniform$(0, 1)$.

What must a PRNG ensure?

To be mathematically valid for simulation, the algorithm must guarantee:

- **Uniformity:** The numbers must be spread evenly across $[0, 1]$. No sub-interval should be favored.
- **Independence:** Successive numbers must lack correlation. Knowing U_n should give you no predictable information about U_{n+1}.
- **Long Period:** Because the system has finite memory and it is determeinistic, the sequence will eventually repeat. The length of this cycle (the "period") must be massive to avoid repeating patterns during a simulation.
- **Efficiency:** It must be computationally fast.

A Simple Algorithm: The Linear Congruential Generator

One of the oldest and most fundamental PRNG algorithms is the **Linear Congruential Generator (LCG)**. It generates a sequence of pseudo-random integers $X_0, X_1, X_2, \ldots$ using a simple recursive formula:

$$X_{n+1} = (aX_n + c) \pmod{m}$$

Where the algorithm is defined by four integer parameters:

- $m > 0$: the **modulus** (which dictates the maximum possible period)
- $a \in \{1, \ldots, m-1\}$: the **multiplier**
- $c \in \{0, \ldots, m-1\}$: the **increment**
- $X_0 \in \{0, \ldots, m-1\}$: the **seed** (the initial starting state)

To obtain the standard uniform output $U \sim \text{Uniform}(0, 1)$, we scale the result:

$$U_{n+1} = \frac{X_{n+1}}{m}$$

LCG: A "Toy" Example

Let's build a miniature LCG to see the algorithm in action.

The Setup

Let Modulus $m = 9$, Multiplier $a = 2$, Increment $c = 0$, and Seed $X_0 = 1$. The recurrence relation is: $X_{n+1} = (2X_n) \pmod 9$

- $X_1 = (2 \times 1) \pmod 9 = 2 \implies \mathbf{U_1 = 2/9 \approx 0.222}$
- $X_2 = (2 \times 2) \pmod 9 = 4 \implies \mathbf{U_2 = 4/9 \approx 0.444}$
- $X_3 = (2 \times 4) \pmod 9 = 8 \implies \mathbf{U_3 = 8/9 \approx 0.889}$
- $X_4 = (2 \times 8) \pmod 9 = 7 \implies \mathbf{U_4 = 7/9 \approx 0.778}$
- $X_5 = (2 \times 7) \pmod 9 = 5 \implies \mathbf{U_5 = 5/9 \approx 0.556}$
- $X_6 = (2 \times 5) \pmod 9 = 1 \implies \mathbf{U_6 = 1/9 \approx 0.111}$

Note: Modern algorithms use highly complex bitwise operations and massive prime moduli to achieve periods like $2^{19937} - 1$, satisfying the strict criteria from our first slide.

Simulating General Uniform Distributions

How do we get from $U \sim \text{Uniform}(0, 1)$ to other uniform distributions? We use simple linear transformations.

1. Continuous Uniform on $[a, b]$

To generate $X \sim \text{Uniform}(a, b)$, we scale U by the width of our target interval $(b - a)$, and then shift it by the starting point a:

$$X = a + (b - a)U$$

2. Discrete Uniform on $\{a, a + 1, \ldots, b\}$

To generate a random integer uniformly, we first create a continuous uniform variable over a slightly wider range $[a, b + 1)$, and then apply the floor function $\lfloor \cdot \rfloor$ to round down to the nearest integer:

$$X = \lfloor a + (b - a + 1)U \rfloor$$

Notice that the multiplier $(b - a + 1)$ is exactly N, the total number of distinct outcomes.

Generating from any CDF: The Inverse Transform Method

What if we want to simulate a random variable X from a distribution with a known Cumulative Distribution Function (CDF), $F(x)$?

The Generalized Inverse Transform Theorem

A valid CDF is non-decreasing but may have flat regions where a standard inverse is undefined. Instead, we use the **generalized (left) inverse**:

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$$

Let $U \sim \text{Uniform}(0, 1)$. If we define a new random variable as:

$$X = F^{-1}(U)$$

Then X is distributed exactly according to $F(x)$.

Generating from any CDF: The Inverse Transform Method

Proof: We want to show that the CDF of our new variable X is $F(x)$:

$$P(X \leq x) = P(F^{-1}(U) \leq x)$$

Because $F(x)$ is non-decreasing and right-continuous, the generalized inverse satisfies the fundamental equivalence $F^{-1}(u) \leq x \iff u \leq F(x)$. Therefore, we can rewrite the probability:

$$P(F^{-1}(U) \leq x) = P(U \leq F(x))$$

Since $U \sim \text{Uniform}(0, 1)$, we know $P(U \leq c) = c$ for any constant $c \in [0, 1]$. Because a valid CDF guarantees $0 \leq F(x) \leq 1$, it follows that:

$$P(U \leq F(x)) = F(x) \quad \blacksquare$$

Inverse Transform Method: Example

Example: The Exponential Distribution

Suppose X has CDF: $F(x) = 1 - e^{-\lambda x}, \quad \text{for } x \geq 0$

1. Set a standard uniform variable U equal to the CDF:

$$U = 1 - e^{-\lambda X}$$

2. Solve algebraically for X to find the inverse $F^{-1}(U)$:

$$e^{-\lambda X} = 1 - U \implies -\lambda X = \ln(1 - U) \implies X = -\frac{1}{\lambda} \ln(1 - U)$$

3. Because $U \sim \text{Uniform}(0, 1)$, the quantity $(1 - U)$ is also exactly distributed as Uniform$(0, 1)$. For computational efficiency, we often simplify the algorithm to:

$$X = -\frac{1}{\lambda} \ln(U)$$

Inverse Transform Method: Discrete Example

Example: A Generic Discrete Distribution

Suppose X takes positive integer values $i = 1, 2, 3, \ldots$ with probabilities $p_i = P(X = i)$. The CDF is a step function:

$$F(k) = P(X \leq k) = \sum_{i=1}^{k} p_i$$

Because the CDF is a step function, we rely entirely on the generalized inverse $F^{-1}(U) = \inf\{x : F(x) \geq U\}$. This essentially finds the first "step" of the CDF that reaches or exceeds U.

1. Generate a standard uniform variable $U \sim \text{Uniform}(0, 1)$.
2. Accumulate the probabilities $p_1, p_2, \ldots$ until the sum is greater than or equal to U. In other words, find the smallest integer k such that:

$$U \leq \sum_{i=1}^{k} p_i$$

Inverse Transform Method: Discrete Example

A Concrete Discrete Distribution

Suppose X takes three values with the following probabilities (p_i) and CDF values $F(x) = P(X \leq x)$:

- $P(X = 1) = 0.2 \implies F(1) = 0.2$
- $P(X = 2) = 0.5 \implies F(2) = 0.2 + 0.5 = 0.7$
- $P(X = 3) = 0.3 \implies F(3) = 0.7 + 0.3 = 1.0$

Finding the Generalized Inverse $F^{-1}(u)$

Recall $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. To find the inverse, we see where a given probability $u \in (0, 1]$ lands in our CDF steps:

$$F^{-1}(u) = \begin{cases} 1 & \text{if } 0 < u \leq 0.2 \\ 2 & \text{if } 0.2 < u \leq 0.7 \\ 3 & \text{if } 0.7 < u \leq 1.0 \end{cases}$$

The Bernoulli Trial

The Bernoulli Trial

A Bernoulli trial is a random experiment with only two possible outcomes, typically labeled as "Success" (1) and "Failure" (0).

- Let p be the probability of success, where $0 \leq p \leq 1$.
- The probability of failure is therefore $1 - p$ (often denoted as q).

Examples:

- Flipping a biased coin (Heads $= 1$, Tails $= 0$).
- Testing a manufactured component (Pass $= 1$, Defective $= 0$).
- A bit transmitted over a noisy channel (Received correctly $= 1$, Error $= 0$).

The Bernoulli Distribution

If a random variable X represents the outcome of a Bernoulli trial, we write $X \sim \text{Bernoulli}(p)$.
The PMF can be compactly written for $x \in \{0, 1\}$ as:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Using our fundamental definitions of expectation and variance:

- **Expectation:**

$$E[X] = \sum_{x \in \{0,1\}} x \cdot P(X = x) = (1 \cdot p) + (0 \cdot (1 - p)) = p$$

- **Second Moment:**

$$E[X^2] = (1^2 \cdot p) + (0^2 \cdot (1 - p)) = p$$

- **Variance:**

$$Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

Standard Discrete Models: The Binomial Distribution

What happens if we repeat a Bernoulli trial multiple times?

Definition via Summation

Let $X_1, X_2, \ldots, X_n$ be n **independent** random variables, where each $X_i \sim \text{Bernoulli}(p)$.
We define a new random variable Y as their sum:

$$Y = \sum_{i=1}^{n} X_i$$

Y represents the total number of "Successes" in n independent trials. We say $Y \sim \text{Binomial}(n, p)$.

Examples:

- The number of Heads in 10 coin flips.
- The number of defective items in a batch of 50.

Binomial Probability Mass Function

To find the PMF of Y, we want the probability of getting exactly k successes (where $k \in \{0, 1, \ldots, n\}$).

Deriving the Probability

1. **Specific Sequence:** The probability of getting k successes followed exactly by $n - k$ failures is $p^k(1-p)^{n-k}$.
2. **Counting Combinations:** There are many ways to arrange k successes among n trials. The number of such unique sequences is given by the binomial coefficient $\binom{n}{k}$.

Multiplying these together yields the PMF:

$$P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Binomial Mean and Variance

We could calculate the expectation by solving $\sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$, which requires complex algebra.
Instead, we use the linear properties we proved earlier!

1. Expectation (Linearity)

Because expectation is a linear operator:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$$

2. Variance (Independence)

Because the trials X_i are **independent**, the covariance terms vanish, allowing us to distribute the variance operator:

$$Var(Y) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

Binomial Expectation:

Let $Y \sim \text{Binomial}(n, p)$ and $q = 1 - p$. We calculate $E[Y]$ directly:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^{n} k \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

Cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[Y] = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k}$$

Factor out np to reconstruct a new binomial coefficient:

$$E[Y] = np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)}$$

Let $j = k - 1$. By the Binomial Theorem, the remaining sum is $(p+q)^{n-1} = 1$:

$$E[Y] = np \cdot 1 = np \quad \blacksquare$$

Binomial Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[Y(Y-1)]$:

$$E[Y(Y-1)] = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k q^{n-k}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting at $k = 2$, cancel $k(k-1)$ with $k!$:

$$E[Y(Y-1)] = \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k}$$

Factor out $n(n-1)p^2$ to rebuild the combinations:

$$E[Y(Y-1)] = n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k}$$

Just as before, the remaining sum is exactly $(p+q)^{n-2} = 1$. Thus:

$$E[Y(Y-1)] = n(n-1)p^2$$

Binomial Variance:

We just found $E[Y(Y-1)] = n(n-1)p^2$ and we know $E[Y] = np$.

1. Find the Second Moment $E[Y^2]$

Expand the factorial moment: $E[Y(Y-1)] = E[Y^2 - Y] = E[Y^2] - E[Y]$.

Rearranging gives:

$$\begin{aligned} E[Y^2] &= E[Y(Y-1)] + E[Y] \\ &= n(n-1)p^2 + np \\ &= n^2p^2 - np^2 + np \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} Var(Y) &= (n^2p^2 - np^2 + np) - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 = np(1-p) \quad \blacksquare \end{aligned}$$

The Sample Proportion ($\hat{p}$)

Instead of looking at the total successes Y, let's examine the **average number of successes** per trial, known as the sample proportion: $\hat{p} = \frac{Y}{n}$.

1. Expectation

$$E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{1}{n}(np) = p$$

The expected proportion of successes is exactly the probability of a single success.

2. Variance

Recall that pulling a constant out of a variance operator squares the constant:

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2}Var(Y) = \frac{1}{n^2}np(1-p) = \frac{p(1-p)}{n}$$

The Sample Proportion: As $n \to \infty$

We just established that the variance of the sample proportion is inversely proportional to the number of trials:

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

The Infinite Limit

What happens mathematically if we repeat the trial an infinite number of times?

$$\lim_{n \to \infty} Var(\hat{p}) = \lim_{n \to \infty} \frac{p(1-p)}{n} = 0$$

Physical Implication: As n grows indefinitely, the sample proportion $\hat{p}$ stops fluctuating and collapses precisely onto the true, underlying probability p. This validates the frequentist definition of probability.

Sums of Independent Random Variables

Let X and Y be two **independent** random variables defined on the same probability space. We are interested in finding the probability distribution of their sum:

$$Z = X + Y$$

To determine the distribution of Z, we analyze all possible pairs (x, y) such that $x + y = z$. Because Y is constrained by $Y = z - X$, independence allows us to determine the distribution using the mathematical operation of **convolution**.

The Discrete Case: PMF Convolution

If X and Y are discrete independent random variables with PMFs $p_X(x)$ and $p_Y(y)$, the PMF of Z is obtained by summing over all valid disjoint supports:

$$p_Z(z) = P(X + Y = z) = \sum_x p_X(x) p_Y(z - x)$$

This operation is frequently denoted as $p_Z = p_X * p_Y$.

Convolution of Continuous Random Variables

The Continuous Case: PDF Convolution

If X and Y are continuous independent random variables characterized by the PDFs $f_X(x)$ and $f_Y(y)$, the density of their sum Z is evaluated via integration:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

Property: Commutativity

Because algebraic addition is commutative ($X + Y = Y + X$), the convolution operator is symmetric. Thus, the order of the functions under the integral can be inverted:

$$\int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y)\, dy$$

Example: Discrete Support (Geometric Sum)

Problem Setup

Let X and Y be independent Geometric random variables representing the number of trials until the first success, both with success probability p. Their PMF for $k \in \{1, 2, 3, \dots\}$ is:

$$p_X(k) = p_Y(k) = (1-p)^{k-1}p$$

Find the PMF of $Z = X + Y$.

Applying the Convolution Formula

Since the minimum value for both X and Y is 1, the minimum value for Z is $1 + 1 = 2$.

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x)$$

The lower limit for x is 1. For $p_Y(z-x)$ to be non-zero, we also need $z - x \geq 1 \implies x \leq z - 1$.

$$\sum_{x=1}^{z-1}$$

Example: Geometric Sum (Solution)

Simplifying the Summation

We combine the terms with base $(1-p)$ and group the p terms:

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{x-1+z-x-1}$$

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{z-2}$$

Final Result

Since the term inside the sum does not depend on x, we are simply adding a constant to itself $z-1$ times:

$$p_Z(z) = (z-1)p^2(1-p)^{z-2}, \quad \text{for } z = 2, 3, 4, \ldots$$

Note: This is exactly the Negative Binomial distribution $NB(2, p)$.

The "Rare Event" Limit

What happens to a Binomial distribution when the number of trials n becomes extremely large, but the probability of success p is extremely small?

The Setup

Consider modeling rare events in a massive population, such as:

- The number of typos in a 500-page book.
- The number of radioactive decays in a millisecond.

Let us take the limit as $n \to \infty$ and $p \to 0$. To keep the mathematics meaningful, we hold their product constant: let $np = \lambda$.
Here, λ represents the **average rate** of successes.

Deriving the Poisson PMF: Setup

Let $Y \sim \text{Binomial}(n, p)$. We substitute $p = \frac{\lambda}{n}$ into the Binomial Probability Mass Function and take the limit as $n \to \infty$:

$$\lim_{n\to\infty} P(Y = k) = \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \lim_{n\to\infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can group the constants (with respect to n) and pull them to the front:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \left[\frac{n!}{(n-k)!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \right]$$

Deriving the Poisson PMF: Evaluation

Let's break the expression inside the limit into three distinct components:

$$= \frac{\lambda^k}{k!} \lim_{n \to \infty} \underbrace{\left[\frac{n(n-1)\dots(n-k+1)}{n^k} \right]}_{\text{Part A}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^n}_{\text{Part B}} \underbrace{\left(1 - \frac{\lambda}{n}\right)^{-k}}_{\text{Part C}}$$

1. **Part A:** There are exactly k terms in the numerator polynomial and n^k in the denominator. As $n \to \infty$, the highest degree terms dominate, and this fraction approaches 1.
2. **Part B:** From standard calculus, the limit definition of the exponential function dictates that $\lim_{n \to \infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$.
3. **Part C:** For any finite k, as $n \to \infty$, the term inside the parentheses approaches 1. Thus, $1^{-k} = 1$.

The Poisson Distribution

Multiplying our evaluated limits together $(1 \cdot e^{-\lambda} \cdot 1)$, we arrive at the PMF for a new, fundamental distribution.

The Poisson PMF

A random variable X follows a Poisson distribution with rate parameter $\lambda > 0$, denoted $X \sim \text{Poisson}(\lambda)$, if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \ldots$$

Mean and Variance of the Poisson: Intuitive

Because we derived this by holding $np = \lambda$ constant:

- **Mean:** $E[X] = \lim_{n\to\infty}(np) = \lambda$
- **Variance:** $Var(X) = \lim_{n\to\infty} np(1-p) = \lambda(1-0) = \lambda$

Poisson Expectation: Direct Derivation

Let $X \sim \text{Poisson}(\lambda)$. By definition, the expected value is:

$$E[X] = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} k\frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is exactly zero, so we can start the sum at $k = 1$. We cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[X] = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!}$$

We factor out $\lambda e^{-\lambda}$ to shift the index. Let $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion for $e^{\lambda} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$:

$$E[X] = \lambda e^{-\lambda}(e^{\lambda}) = \lambda \quad \blacksquare$$

Poisson Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting the sum at $k = 2$, we cancel $k(k-1)$ with the $k!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!}$$

Factor out $\lambda^2 e^{-\lambda}$ and let $j = k - 2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Once again, the infinite sum resolves exactly to e^{λ}:

$$E[X(X-1)] = \lambda^2 e^{-\lambda}(e^{\lambda}) = \lambda^2$$

Poisson Variance: Direct Derivation

We now have the two necessary components: $E[X(X-1)] = \lambda^2$ and $E[X] = \lambda$.

1. Find the Second Moment $E[X^2]$

Expand the factorial moment: $E[X(X-1)] = E[X^2] - E[X]$.

Rearranging gives us:

$$\begin{aligned} E[X^2] &= E[X(X-1)] + E[X] \\ &= \lambda^2 + \lambda \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} Var(X) &= (\lambda^2 + \lambda) - (\lambda)^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \quad \blacksquare \end{aligned}$$

Waiting Time: The Geometric Distribution

We return to our sequence of independent Bernoulli trials (where $P(\text{Success}) = p$ and $P(\text{Failure}) = q = 1 - p$).
Instead of counting the *number* of successes in a fixed time, we now ask:
How long must we wait until the very first success?

Defining the Random Variable

Let X be the number of trials needed to get the first success.

- The only way for $X = k$ to occur is to have exactly $k - 1$ failures followed immediately by 1 success.
- Because the trials are independent, we multiply their probabilities.

The Geometric PMF

We say $X \sim \text{Geometric}(p)$. Its Probability Mass Function is:

$$P(X = k) = q^{k-1}p, \quad \text{for } k = 1, 2, 3, \ldots$$

Geometric Expectation:

By definition, $E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p$.
We can pull the constant p outside the sum:

$$E[X] = p\sum_{k=1}^{\infty} kq^{k-1}$$

To evaluate this infinite series, we recall the standard geometric series for $|q| < 1$:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

If we take the derivative of both sides with respect to q, the k drops down:

$$\frac{d}{dq}\sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq}(1-q)^{-1} = \frac{1}{(1-q)^2}$$

Substituting this result back into our expectation formula:

$$E[X] = p\left[\frac{1}{(1-q)^2}\right] = p\left[\frac{1}{p^2}\right] = \frac{1}{p} \quad \blacksquare$$

Geometric Variance and Intuition

To find the variance, we would use the second derivative of the geometric series to find the second factorial moment $E[X(X-1)]$.

Moments Summary

For $X \sim \text{Geometric}(p)$:

- **Mean:** $E[X] = \frac{1}{p}$
- **Variance:** $Var(X) = \frac{1-p}{p^2}$

Physical Intuition

The expected value $E[X] = 1/p$ aligns perfectly with human intuition:

- If you flip a fair coin ($p = 1/2$), you expect to wait $1/(1/2) = 2$ flips to see the first Heads.
- If you roll a fair die looking for a six ($p = 1/6$), you expect to wait $1/(1/6) = 6$ rolls to see the first 6.

The Memoryless Property

A unique and defining characteristic of the Geometric distribution is that it is **memoryless**.

Formal Definition

For any integers $s > 0$ and $t > 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

The probability of waiting t additional trials for a success, given that we have already waited s trials, is exactly the same as the probability of waiting t trials from the very beginning.

The samples do not "remember" past failures. Suppose you are rolling a fair die waiting for a 6.

- You have already rolled 10 times and failed.
- The die is not "due" for a 6. The probability that it takes 3 *more* rolls is exactly the same as taking 3 rolls on a brand new die.

Proof of the Memoryless Property

Prerequisite: Check that $P(X > k) = q^k$. (To fail to get a success in the first k trials means getting k failures in a row).

Using the definition of conditional probability:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)}$$

Since the event $\{X > s + t\}$ is a subset of the event $\{X > s\}$, their intersection is simply $\{X > s + t\}$. We substitute our tail probabilities:

$$\begin{aligned} P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{q^{s+t}}{q^s} \\ &= q^t \\ &= P(X > t) \quad \blacksquare \end{aligned}$$

Conclusion: The past history of failures cancels out.

Generalizing the Wait: Negative Binomial

We just looked at the waiting time for the first success. Now, we generalize: **How long must we wait until the r-th success?**

Defining the Random Variable

Let X be the number of trials needed to achieve exactly r successes.

- The Geometric distribution is simply a special case where $r = 1$.
- For $X = k$ to occur, the very last trial (trial k) must be a success.
- The previous $k - 1$ trials must contain exactly $r - 1$ successes.

The Negative Binomial PMF

Let's build the Probability Mass Function for $X \sim \text{NB}(r, p)$.

Deriving the Formula

- Probability of exactly $r-1$ successes in the first $k-1$ trials (Binomial): $\binom{k-1}{r-1} p^{r-1} q^{(k-1)-(r-1)}$
- Probability that the k-th trial is a success: p
- Multiplying these independent events gives us our PMF!

The PMF

$$P(X = k) = \binom{k-1}{r-1} p^r q^{k-r}$$

for $k = r, r+1, r+2, \ldots$

Negative Binomial Moments

We could use calculus tricks to find the expectation, but there is a much more elegant way using linearity of expectation!

Sum of Independent Geometrics

We can think of waiting for r successes as waiting for the first success, then waiting for the second, and so on.

$$X = Y_1 + Y_2 + \cdots + Y_r$$

where each $Y_i \sim \text{Geometric}(p)$.

Moments Summary

Because expectation is linear and the wait times are independent:

- **Mean:** $E[X] = r \cdot E[Y_i] = \frac{r}{p}$
- **Variance:** $Var(X) = r \cdot Var(Y_i) = \frac{r(1-p)}{p^2}$

The Fish Population Problem

We want to estimate the total number of fish in a lake, but we obviously cannot drain the lake to count them all. Instead, we use a statistical inference technique called **Capture-Recapture**.

The Setup

- **Step 1 (Tagging):** We capture a specific number of fish (let's call this K), tag them securely, and release them back into the water to mix with the general population.
- **Step 2 (Sampling):** After giving them time to disperse, we capture a new sample of n fish from the lake.

The Random Variable

Let X be the number of tagged fish that end up in our second sample. We want to use X to estimate the total, unknown population size, N.

The Hypergeometric PMF

Because we are pulling fish from a finite lake and not throwing them back between individual catches, the trials are **not independent**. The Binomial distribution assumes replacement; the **Hypergeometric** handles scenarios *without* replacement.

The Parameters

- N: Total population size (unknown)
- K: Total tagged fish in the lake
- n: Our sample size (number of draws)
- k: The observed number of tagged fish in our sample

To find the probability of catching exactly k tagged fish, we divide the number of successful combinations by the total possible combinations:

$$P(X = k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Estimating the Total Population

Using Expectation for Inference

The expected value of a Hypergeometric random variable relies on the proportion of "successes" in the overall population:

$$E[X] = n\frac{K}{N}$$

- We use our actual observed number of tagged fish, k, as our best estimate for the expected value $E[X]$.

Solving for N: By substituting k in place of the expectation, we can rearrange the algebra to estimate the total population size:

$$N \approx \frac{n \cdot K}{k}$$

The Exponential Distribution

Definition

A continuous random variable X follows an exponential distribution with rate parameter $\lambda > 0$ if its probability density function (PDF) is:

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x \geq 0 \\ 0 & \text{for } x < 0 \end{cases}$$

The CDF

$$\begin{aligned} F(x) &= \int_{-\infty}^{x} f(t)\, dt \\ &= \int_{0}^{x} \lambda e^{-\lambda t}\, dt \\ &= -e^{-\lambda x} - \left(-e^{0}\right) \\ &= 1 - e^{-\lambda x} \end{aligned}$$

Exponential Distribution: Mean Derivation

Let $X \sim \text{Exponential}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$. By definition:

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx$$

We use Integration by Parts (IBP): $\int u\, dv = uv - \int v\, du$.

- Let $u = x \implies du = dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X] &= \left[-xe^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) dx \\ &= (0 - 0) + \int_0^\infty e^{-\lambda x} dx \quad \left(\text{Note: } \lim_{x\to\infty} \frac{x}{e^{\lambda x}} = 0 \text{ by L'Hôpital's}\right) \\ &= \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty \\ &= 0 - \left(-\frac{1}{\lambda}\right) = \frac{1}{\lambda} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Second Moment

To find the variance, we first need $E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$. Using IBP again:

- Let $u = x^2 \implies du = 2x\,dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$E[X^2] = \left[-x^2 e^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) \cdot 2x\,dx$$
$$= 0 + 2\int_0^\infty x e^{-\lambda x} dx$$

Notice that the remaining integral is almost $E[X]$. We multiply and divide by λ:

$$E[X^2] = \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx$$
$$= \frac{2}{\lambda} \cdot E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}$$

Exponential Distribution: Variance

Now we apply the variance formula:

$$Var(X) = E[X^2] - (E[X])^2$$

The Final Calculation

Substituting our derived moments:

$$\begin{aligned} Var(X) &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda^2} \quad \blacksquare \end{aligned}$$

Summary for $X \sim \text{Exp}(\lambda)$

- **Mean:** $E[X] = 1/\lambda$
- **Standard Deviation:** $SD(X) = 1/\lambda$

The Normal Distribution

Definition

The Normal (Gaussian) Distribution A continuous random variable X follows a Normal distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$, denoted as $X \sim N(\mu, \sigma^2)$, if its Probability Density Function (PDF) is given by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{for } -\infty < x < \infty$$

Role of the Parameters

- **Mean (μ):** The location parameter. It dictates the center of symmetry.
- **Standard Deviation (σ):** The scale parameter. It dictates the width and height of the bell curve.

Proving the Normal Density Integrates to 1

We want to show that $\int_{-\infty}^{\infty} f(x)dx = 1$. Let's look at the standard kernel integral. Because the integrand is perfectly symmetric (even), we can write it as twice the integral over the positive domain:

$$I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = 2 \int_{0}^{\infty} e^{-x^2/2} dx$$

We apply the substitution $u = \frac{x^2}{2} \implies x = \sqrt{2}u^{1/2}$, which gives $dx = \frac{1}{\sqrt{2}} u^{-1/2} du$. Substituting these into the integral maps it directly to the Gamma distribution framework:

$$I = 2 \int_{0}^{\infty} e^{-u} \left(\frac{1}{\sqrt{2}} u^{-1/2} du \right) = \sqrt{2} \int_{0}^{\infty} u^{-1/2} e^{-u} du$$

Proving the Normal Density Integrates to 1

Recalling the Gamma Function

Recall that $\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u} du$. Matching our exponent, $\alpha - 1 = -1/2 \implies \alpha = 1/2$.

Knowing the standard result that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$:

$$I = \sqrt{2} \cdot \Gamma\left(\frac{1}{2}\right) = \sqrt{2} \cdot \sqrt{\pi} = \sqrt{2\pi}$$

Applying the Normalization Constant

Using the standard transform $z = \frac{x-\mu}{\sigma} \implies dx = \sigma dz$:

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} (\sigma dz) = \frac{\sqrt{2\pi}}{\sqrt{2\pi}} = 1 \quad \blacksquare$$

Deriving the Mean: $E[X] = \mu$

By definition, the expected value of $X \sim N(\mu, \sigma^2)$ is:

$$E[X] = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

We apply the standard linear substitution $z = \frac{x-\mu}{\sigma} \implies x = \mu + \sigma z$ and $dx = \sigma dz$:

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} (\mu + \sigma z) \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-z^2/2} (\sigma dz) \\ &= \mu \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz}_{\text{Integrates to 1}} + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2} dz \end{aligned}$$

- The first term evaluates simply to $\mu \cdot 1 = \mu$.
- The integrand in the second term, $g(z) = ze^{-z^2/2}$, is an **odd function** ($g(-z) = -g(z)$). Integrating any odd function over symmetric limits $(-\infty, \infty)$ yields exactly 0. Hence $E[X] = \mu + 0 = \mu$ ■

Deriving the Variance:

Using the standard substitution $z = \frac{x-\mu}{\sigma} \implies dx = \sigma dz$, the variance simplifies to:

$$Var(X) = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx = \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} dz = \frac{2\sigma^2}{\sqrt{2\pi}} \int_{0}^{\infty} z^2 e^{-z^2/2} dz$$

Substitute $u = \frac{z^2}{2} \implies z = \sqrt{2}u^{1/2}$ and $dz = \frac{1}{\sqrt{2}} u^{-1/2} du$:

$$2 \int_0^{\infty} (2u) e^{-u} \left(\frac{1}{\sqrt{2}} u^{-1/2} du \right) = 2\sqrt{2} \int_0^{\infty} u^{1/2} e^{-u} du$$

We recognize the Gamma function $\int_0^{\infty} u^{\alpha-1} e^{-u} du$. Here, $\alpha = 3/2$.

Using the property $\Gamma(n+1) = n\Gamma(n)$, we know $\Gamma(3/2) = \frac{1}{2}\Gamma(1/2) = \frac{\sqrt{\pi}}{2}$:

$$2\sqrt{2} \cdot \Gamma\left(\frac{3}{2}\right) = 2\sqrt{2} \left(\frac{\sqrt{\pi}}{2} \right) = \sqrt{2\pi}$$

Substituting back: $Var(X) = \frac{\sigma^2}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = \sigma^2$ ■

The Mode of the Normal Distribution

The **mode** is the value of x that maximizes the density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Minimizing the Penalty

To maximize the exponential function e^{-y}, we must **minimize** the positive term y in the exponent:

$$y = \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2$$

- Because y is a squared term, its absolute minimum is exactly 0.
- This minimum occurs strictly when $x - \mu = 0 \implies x = \mu$.

Conclusion: The distance penalty is zero exactly at the mean, meaning the density peaks exactly at the mean.

$$\therefore \text{Mode} = \mu \quad \blacksquare$$

The Median of the Normal Distribution

The Geometric Argument

We can find the median by looking at the probability density function:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

- The distance term in the exponent, $(x-\mu)^2$, squares the deviation. This means any point c units below the mean has the exact same density as a point c units above the mean: $f(\mu - c) = f(\mu + c)$. This creates a curve that is a perfect mirror image of itself, strictly centered at $x = \mu$.

Conclusion

Because the total area under the curve is exactly 1, and the curve is perfectly symmetric around μ, the area to the left of μ must equal the area to the right of μ.

$$P(X \leq \mu) = P(X \geq \mu) = 0.5 \implies \text{Median} = \mu \quad \blacksquare$$

Linear Transformations of Normal Variables

Theorem

An important fact about normal random variables is that if X is normally distributed with parameters μ and σ^2, then $Y = aX + b$ is normally distributed with parameters $a\mu + b$ and $a^2\sigma^2$.

Proof via CDF

To prove this statement, suppose that $a > 0$. (The proof when $a < 0$ is similar.) Let F_Y denote the cumulative distribution function of Y. Then

$$\begin{aligned} F_Y(x) &= P\{Y \leq x\} \\ &= P\{aX + b \leq x\} \\ &= P\left\{X \leq \frac{x-b}{a}\right\} \\ &= F_X\left(\frac{x-b}{a}\right) \end{aligned}$$

where F_X is the cumulative distribution function of X.

Linear Transformations of Normal Variables

Density Function Calculation

By differentiation, the density function of Y is then

$$\begin{aligned} f_Y(x) &= \frac{1}{a} f_X\left(\frac{x-b}{a}\right) \\ &= \frac{1}{\sqrt{2\pi}a\sigma} \exp\left\{-\left(\frac{x-b}{a}-\mu\right)^2/2\sigma^2\right\} \\ &= \frac{1}{\sqrt{2\pi}a\sigma} \exp\left\{-(x-b-a\mu)^2/2(a\sigma)^2\right\} \end{aligned}$$

Conclusion

which shows that Y is normal with parameters $a\mu + b$ and $a^2\sigma^2$. ■

Computing Normal Probabilities

The Standard Normal Functions

The cumulative distribution function (CDF) of a normal random variable cannot be expressed in a simple closed form and is typically calculated numerically. For the standard normal distribution $Z \sim \mathcal{N}(0, 1)$, the CDF and density (PDF) are given special notation:

- **Standard CDF:** $\Phi(x) = P(Z \leq x) = \int_{-\infty}^{x} \phi(t)\, dt$
- **Standard PDF:** $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

General Normal Variables

Thanks to the linear transformation properties established earlier, the CDF for any general normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ can be evaluated using the standard function Φ:

$$P(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

Population and Sample

- **Population**: entire set of interest
- **Sample**: subset drawn from population

We make inference about the population using samples.
Inference is typically based on a **statistic** computed from the sample, which is used to draw conclusions about the population.
Assume:

- Each observation is drawn randomly
- Observations are independent

$$X_1, X_2, \ldots, X_n \text{ i.i.d.}$$

Statistics and Parameters

A **parameter** is a fixed, usually unknown numerical constant that describes a population distribution.
Example: If $X \sim N(\mu, \sigma^2)$. μ and σ^2 are the parameters of the distribution.

A **statistic** is any function of the data.
Examples of statistics:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

$$X_{(n)} = \max_i X_i$$

Due to the randomness of the drawn samples, statistics are random variables, whereas parameters are fixed constants.

Example of Sampling distribution of a statistic using Bernoulli Distribution

A Bernoulli random variable:

$$X = \begin{cases} 1 & \textit{with prob } p \\ 0 & \textit{with prob } 1-p \end{cases}$$

$$\mathbb{E}[X] = p$$

$$\mathrm{Var}(X) = p(1-p)$$

Used to model:

- Yes / No outcomes
- Success / Failure experiments
- coin toss

Statistic for Bernoulli Data

Let

$$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$$

Sample proportion:

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{S_n}{n}, \quad \text{where } S_n = \sum_{i=1}^{n} X_i$$

- $S_n \sim \text{Binomial}(n, p)$
- $\hat{p}$ is a scaled Binomial random variable

Exact distribution of $\hat{p}$:

$$\mathbb{P}\left(\hat{p} = \frac{k}{n}\right) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

Sampling Distribution of Sample Proportion

For Bernoulli data, the sampling distribution of the statistic $\hat{p}$ is **known exactly** for any sample size n.

Let

$$S_n = \sum_{i=1}^{n} X_i \sim \text{Binomial}(n, p), \quad \hat{p} = \frac{S_n}{n}$$

From properties of the Binomial distribution:

$$\mathbb{E}[S_n] = np, \qquad \text{Var}(S_n) = np(1-p)$$

Hence,

$$\mathbb{E}[\hat{p}] = p, \qquad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}$$

- The exact sampling distribution of $\hat{p}$ is fully known
- No approximation is required in this case

Example of Sampling Distribution using Binomial Data

A Binomial random variable:

$$Y \sim \text{Binomial}(n, p)$$

$$\mathbb{P}(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n$$

$$\mathbb{E}[Y] = np, \qquad \text{Var}(Y) = np(1-p)$$

Used to model:

- Number of successes in n trials
- Aggregated Bernoulli outcomes
- Repeated experiments with fixed n

Statistic for Binomial Data

Let

$$Y_1, \ldots, Y_m \sim \text{Binomial}(n, p)$$

Sample mean:

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

- $\bar{Y}$ is the average number of successes
- $\sum_{i=1}^{m} Y_i \sim \text{Binomial}(mn, p)$

Exact distribution of $\bar{Y}$:

$$\mathbb{P}\left(\bar{Y} = \frac{k}{m}\right) = \binom{mn}{k} p^k (1-p)^{mn-k}, \quad k = 0, 1, \ldots, mn$$

Sampling Distribution of Sample Mean

For Binomial data, the sampling distribution of the statistic $\bar{Y}$ is **known exactly**.

Let

$$Y_1, \ldots, Y_m \sim \text{Binomial}(n, p), \quad \bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

From properties of the Binomial distribution:

$$\mathbb{E}[\bar{Y}] = np, \qquad \text{Var}(\bar{Y}) = \frac{np(1-p)}{m}$$

- Exact sampling distribution is discrete
- Variance shrinks as sample size m increases

Example of Sampling Distribution using Poisson Data

A Poisson random variable:

$$Y \sim \text{Poisson}(\lambda)$$

$$\mathbb{P}(Y = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

$$\mathbb{E}[Y] = \lambda, \qquad \text{Var}(Y) = \lambda$$

Used to model:

- Number of events in a fixed interval
- Arrival processes (calls, packets, failures)
- Rare event counts

Statistic for Poisson Data

Let

$$Y_1, \ldots, Y_m \sim \text{Poisson}(\lambda)$$

Sample mean:

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

- $\bar{Y}$ is a statistic used to estimate λ
- Distribution of $\bar{Y}$ depends on the sum $\sum Y_i$

Convolution of Two Poisson Variables: Setup

Let

$$Y_1 \sim \text{Poisson}(\lambda_1), \quad Y_2 \sim \text{Poisson}(\lambda_2)$$

be independent.

We want the distribution of

$$S = Y_1 + Y_2$$

For $k = 0, 1, 2, \ldots,$

$$\mathbb{P}(S = k) = \mathbb{P}(Y_1 + Y_2 = k)$$

The event $\{Y_1 + Y_2 = k\}$ occurs when:

$$(Y_1, Y_2) = (0, k), (1, k-1), \ldots, (k, 0)$$

Convolution of Two Poisson Variables

Using independence,

$$\mathbb{P}(Y_1 + Y_2 = k) = \sum_{j=0}^{k} \mathbb{P}(Y_1 = j)\mathbb{P}(Y_2 = k - j)$$

Substitute Poisson pmfs:

$$= \sum_{j=0}^{k} \frac{e^{-\lambda_1}\lambda_1^j}{j!} \frac{e^{-\lambda_2}\lambda_2^{k-j}}{(k-j)!}$$

Simplifying,

$$= \frac{e^{-(\lambda_1+\lambda_2)}}{k!} \sum_{j=0}^{k} \binom{k}{j} \lambda_1^j \lambda_2^{k-j}$$

By the Binomial theorem,

$$\mathbb{P}(Y_1 + Y_2 = k) = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^k}{k!}$$

Sampling Distribution of Sample Mean

Using the convolution result,

$$\sum_{i=1}^{m} Y_i \sim \text{Poisson}(m\lambda)$$

Therefore,

$$\bar{Y} = \frac{1}{m} \sum_{i=1}^{m} Y_i$$

From properties of the Poisson distribution:

$$\mathbb{E}[\bar{Y}] = \lambda, \qquad \text{Var}(\bar{Y}) = \frac{\lambda}{m}$$

- $\bar{Y}$ is an unbiased estimator of λ
- Variability decreases as sample size m increases

Probability as Area: Uniform$(0, \theta)$

For $0 \leq a < b \leq \theta$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b \frac{1}{\theta}\, dx = \frac{b-a}{\theta}$$

- Height of the pdf is constant
- Probability is proportional to interval length

Let

$$X_1, \ldots, X_n \sim \text{Uniform}(0, \theta)$$

The parameter θ is **unknown**.

Natural Statistic: Sample Maximum

Since all observations lie in $[0, \theta]$,

$$\max(X_1, \ldots, X_n) \leq \theta$$

Define the statistic:

$$M_n = \max\{X_1, \ldots, X_n\}$$

- M_n contains information about θ
- Larger samples tend to produce larger M_n

Sampling Distribution of the Sample Maximum

For $0 \leq x \leq \theta$,

$$\mathbb{P}(M_n \leq x) = \mathbb{P}(X_1 \leq x, \ldots, X_n \leq x)$$

Using independence,

$$= \left(\frac{x}{\theta}\right)^n$$

Hence, the CDF of M_n is

$$F_{M_n}(x) = \begin{cases} \left(\frac{x}{\theta}\right)^n, & 0 \leq x \leq \theta \\ 1, & x \geq \theta \end{cases}$$

Using the Statistic to Estimate θ

From the distribution of M_n,

$$\mathbb{E}[M_n] = \frac{n}{n+1}\theta$$

This suggests the estimator:

$$\hat{\theta} = \frac{n+1}{n} M_n$$

- $\hat{\theta}$ is an unbiased estimator of θ
- Accuracy improves as sample size increases

Normal Distribution

A continuous random variable X is said to follow a **Normal**(θ, σ^2) distribution if its pdf is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\theta)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

- θ is the mean (location parameter)
- Variance is fixed and equal to σ^2

$$\mathbb{E}[X] = \theta, \qquad \mathrm{Var}(X) = \sigma^2$$

Statistic for Normal(θ, σ^2) Data

Let

$$X_1, \ldots, X_n \sim \text{Normal}(\theta, \sigma^2)$$

The parameter θ is **unknown**.

- Goal: Estimate θ using the observed data
- Here we have assumed the population variance to be known and equal to σ^2 for simplicity.

Natural Statistic: Sample Mean

Define the sample mean:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

- $\bar{X}$ summarizes the central tendency of the data
- It is a natural candidate to estimate θ

Sampling Distribution of the Sample Mean

Since the Normal distribution is closed under averaging,

$$\bar{X} \sim \text{Normal}\left(\theta, \frac{\sigma^2}{n}\right)$$

Therefore,

$$\mathbb{E}[\bar{X}] = \theta, \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

- $\bar{X}$ is an unbiased estimator of θ
- Variability decreases as n increases

The Law of Large Numbers: Intuition

In most of the examples above, we have seen two key properties regarding the sample mean $\bar{X}$:

- The expected value of the sample mean is exactly the population mean: $E[\bar{X}] = \mu$.
- The variance shrinks as the sample size n increases: $Var(\bar{X}) = \frac{\sigma^2}{n}$.

Notice what happens as the sample size grows arbitrarily large ($n \to \infty$): the variance $\frac{\sigma^2}{n}$ approaches 0. Intuitively, if the variance of $\bar{X}$ approaches 0, the probability that $\bar{X}$ deviates far from μ must also shrink to 0. This brings us to a fundamental concept: the **Law of Large Numbers**.

The Weak Law of Large Numbers (WLLN)

Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables with expected value μ. For any arbitrarily small margin $\epsilon > 0$:

$$\lim_{n \to \infty} P\left(|\bar{X} - \mu| \geq \epsilon\right) = 0$$

The Central Limit Theorem (CLT)

Stabilizing the Variance

The Law of Large Numbers shows that $\bar{X}$ collapses to μ because $Var(\bar{X}) = \frac{\sigma^2}{n} \to 0$. However, if we look at the deviation scaled by $\sqrt{n}$, we get a stable variance:

$$Var(\sqrt{n}(\bar{X} - \mu)) = (\sqrt{n})^2 Var(\bar{X}) = n\left(\frac{\sigma^2}{n}\right) = \sigma^2 \neq 0$$

Because this scaled variance remains constant as $n \to \infty$, the distribution of this scaled variable stabilizes rather than collapsing to a single point.

A Universal Limit

Remarkably, under mild regularity conditions (specifically, having a finite variance), the limiting distribution of this scaled variable is exactly the same **regardless of the underlying population distribution**.

The Central Limit Theorem (CLT)

Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be a sequence of i.i.d. random variables with expected value μ and finite variance $\sigma^2 > 0$. As the sample size $n \to \infty$, the distribution of the standardized sample mean converges to the standard normal distribution:

$$\lim_{n\to\infty} P\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq x\right) = \Phi(x)$$

where $\Phi(x)$ is the cumulative distribution function of $\mathcal{N}(0, 1)$.

The Chi-Square (χ^2) Distribution

Motivation

We have established the behavior of the sample mean $\bar{X}$, but analyzing the spread of our data requires looking at the sample variance. To understand its sampling distribution, we must introduce the Chi-square distribution.

Definition

Let $Z_1, Z_2, \ldots, Z_k$ be independent standard normal random variables, where $Z_i \sim \mathcal{N}(0, 1)$. The sum of their squared values,

$$Q = \sum_{i=1}^{k} Z_i^2$$

is distributed according to the **Chi-square distribution** with k degrees of freedom. We denote this as $Q \sim \chi_k^2$.

Sampling Distribution of the Sample Variance

The Sample Variance

Recall that the sample variance for a sample of size n is defined as:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Distribution for Normal Populations

If the underlying data $X_1, X_2, \ldots, X_n$ are independent and identically distributed draws from a normal population $\mathcal{N}(\mu, \sigma^2)$, the scaled sample variance follows a Chi-square distribution with $n-1$ degrees of freedom:

$$\frac{(n-1)S^2}{\sigma^2} = \sum_{i=1}^{n}\left(\frac{X_i - \bar{X}}{\sigma}\right)^2 \sim \chi^2_{n-1}$$

Mean and Variance of χ^2_k

Because we defined χ^2_k as a sum of Z^2, we do not need to evaluate the dense Gamma integrals to find its moments. We can just use the properties of $Z \sim N(0,1)$.

The Expected Value

Recall the variance formula: $Var(Z) = E[Z^2] - (E[Z])^2$. Since $Var(Z) = 1$ and $E[Z] = 0$, this implies $E[Z^2] = 1$.

$$E[\chi^2_k] = E\left[\sum_{i=1}^{k} Z_i^2\right] = \sum_{i=1}^{k} E[Z_i^2] = \sum_{i=1}^{k}(1) = k \quad \blacksquare$$

The Variance

It is a known property of standard normals that the variance of their square is exactly 2 ($Var(Z^2) = 2$). Since the Z_i are independent:

$$Var(\chi^2_k) = Var\left(\sum_{i=1}^{k} Z_i^2\right) = \sum_{i=1}^{k} Var(Z_i^2) = \sum_{i=1}^{k}(2) = 2k \quad \blacksquare$$

The Probability Density Function of χ^2_k

While we intuitively defined it as a sum of squared normals, the χ^2_k distribution is formally defined by its continuous Probability Density Function.

Definition

Chi-Square PDF Let $X \sim \chi^2_k$. The probability density function of X is:

$$f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-x/2} \quad \text{for } x > 0$$

(and $f(x) = 0$ for $x \leq 0$).

The Chi-Square distribution is actually a specific case of the **Gamma Distribution**!

- Shape parameter: $\alpha = k/2$
- Scale parameter: $\theta = 2$ (or Rate $\lambda = 1/2$)

Exercise: Calculate the mean and variance using the pdf.

Student's t-Distribution

When we use the Central Limit Theorem, we standardize the mean: $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$.

The Problem: In reality, the true population standard deviation σ is almost always unknown. We must estimate it using the sample standard deviation S.

The Formal Definition

Replacing the constant σ with a random variable S changes the math. We define the t-distribution by combining our two foundational distributions.
Let $Z \sim N(0,1)$ and $Q \sim \chi^2_v$ be independent. The ratio:

$$T = \frac{Z}{\sqrt{Q/v}} \sim t_v$$

follows a Student's t-distribution with v degrees of freedom.

The Shape: Uncertainty and "Heavy Tails"

Because T divides a standard normal variable by another random variable (the chi-square component), it introduces an extra layer of uncertainty.

Geometric Impact

The t-distribution is symmetric and centered at 0, just like the Standard Normal. However, this extra variation pushes probability outward from the center:

- The peak is shorter than $N(0, 1)$.
- The tails are **thicker** (heavy tails), meaning extreme values are mathematically more likely to occur than under a normal curve.

Applying the t-Distribution

How does this definition map to our actual sample data?

The t-Statistic

When we standardize a sample mean using the sample standard deviation, the resulting statistic follows a t-distribution:

$$T_n = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

Degrees of Freedom ($n-1$)

The distribution has $v = n - 1$ degrees of freedom, rather than n.

Why? To compute S, we first had to compute $\bar{X}$. We "spent" one independent piece of information to find the center, leaving $n - 1$ independent pieces of information to calculate the variation.